Lab 5: Disk Management and Logical Volume Setup

Objective: Attach a 15GB disk to your VM, partition it into 5GB, 5GB, 3GB, and 2GB sections. Use the first 5GB partition as a file system, configure the 2GB partition as swap, initialize the second 5GB as a Volup Group (VG) with a Logical Volume (LV), then extend the LV by adding the 3GB partition.

1-add disk and divide to partition (5.5.3.2)

-fdisk /dev/sdb

m>>>>n

```
11  fdisk sdb
12  sudo fdisk /dev/sdb
13  df -h
14  lsblk
15  mount /dev/sdb1  /media
16  df -h4
17  df -h
18  ls
19  mount /dev/sdb1  /videos
20  df -h
21  lsblk
22  fdisk  /dev/sdb
23  sudo fdisk /dev/sdb
24  lsblk
25  sudo mkfs.ext4 /dev/sdb1
26  sudo mkswap /dev/sdb4
27  sudo swapon /dev/sdb4
28  sudp pvcreate /dev/sdb2
29  sudo pvcreate /dev/sdb2
30  pvcreate /dev/sdb2
31  sudo apt install lvm
32  sudo apt install lvm2
```

```
loop10   7:10   0  38.7M  1 loop /snap/snapd/21465
loop11   7:11   0   568K  1 loop /snap/snapd-desktop-integration/253
loop12   7:12   0  44.3M  1 loop /snap/snapd/23258
sda      8:0    0    20G  0 disk
├sda1   8:1    0     1M  0 part
└sda2   8:2    0    20G  0 part /var/snap/firefox/common/host-hunspell
                                 /
sdb      8:16   0    15G  0 disk
├sdb1   8:17   0     5G  0 part
├sdb2   8:18   0     5G  0 part
├sdb3   8:19   0     3G  0 part
└sdb4   8:20   0     2G  0 part
sr0     11:0    1  53.7M  0 rom  /media/ahmed/VMware Tools
```

2- **change formate /dev/sdb1 =5G as a file system,**

- Sudo mkfs.ext4 /dev/sdb1

3- **change formate /dev/sdb4 = 2G as swap**

Sudo mkswap /dev/sdb4

Sudo swapon /dev/sdb4

```
sudo mkfs.ext4 /dev/sdb1
sudo mkswap /dev/sdb4
sudo swapon /dev/sdb4
```

4- **change partition to Physical Volume**

- Create Physical Volume

Sudo pvcreate /dev/sdb2

-to display detailed information about the Physical Volumes (PVs) on asystem.

Sudo pvdisplay

-create volume group . name VG from physical volume /dev/sdb2Sudo vgcreate VG /dev/sdb2

-create logical volume . name lo from volume group VG

Sudo lvcreate -L +5G -n lo VG

```
pvcreate /dev/sdb2
sudo pvcreate /dev/sdb2
pvdisplay
sudo pvdisplay
sudo vgcreate VG /dev/sdb2
sudo lvcreate -L  5G -n lo VG
lvdisplay
vdisplay
sudo lvdisplay
lsblk
df -h
```

5- **increase the size of an existing Logical Volume (LV)**

-change partion /dev/sdb3 to physical volumeSudo

pvcreate /dev/sdb3

-add physical volume to volume group

Sudo vgextend VG /dev/sdb3

- add new space to logical volumeSudo

lvextend -L 3G /dev/VG/lo

```
45  sudo vgextend VG /dev/sdb3
46  sudo lvextend -L +3G  /dev/VG/lo
47  sudo lvextend -L +3G  -n lo VG
48  sudo lvextend -L +3G  /dev/VG/lo
49  sudo lvdisplay
50  lvdisplay
51  sudo lvextend -L +3G /dev/VG/lo
52  sudo lvcreate -L +5 log /dev/sdb2
53  sudo lvcreate -L +5 lo /dev/sdb2
54  history
55  sudo lvcreate -L 5G lo VG
56  sudo lvextend -L +3G  /dev/VG/lo
57  sudo vgdisplay
58  sudo lvcreate -L 5G -n lo VG
59  sudo lvextend -L +3G /dev/VG/lo
60  sudo resize2fs /dev/my_volume_group/my_logical_volume
```

Automount by fstab :

```
# / was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/4abcf1e7-e878-42ed-ab50-0173ae01157b / ext4 defaults 0 1
/swap.img       none    swap    sw      0       0
/dev/VG1/lve     /ivolve        ext4   defaults 0 0
/dev/sdc5        none    swap   sw             0 0
/dev/sdc1       /ivolve2        ext4   defaults 0 0
~
~
~
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~
~
~
"/etc/fstab" 13L, 618B                                    13,51         All
```

```
├─sdc1        8:33   0     5G  0 part /ivolve2
├─sdc2        8:34   0     5G  0 part
│ └─VG1-lve 252:1    0     7G  0 lvm  /ivolve
├─sdc3        8:35   0     3G  0 part
│ └─VG1-lve 252:1    0     7G  0 lvm  /ivolve
├─sdc4        8:36   0     1K  0 part
└─sdc5        8:37   0     2G  0 part [SWAP]
```